

sometimes be maintained by hCG alone.

• **Delayed puberty associated with insufficient gonadotropic pituitary function**
1500 IU two to three times weekly for at least six months

• **Cryptorchidism, not due to anatomical obstruction**
- Under 2 years of age: 250 IU twice weekly for six weeks
- Under 6 years of age: 500 - 1000 IU twice weekly for six weeks
- Over 6 years of age: 1500 IU twice weekly for six weeks.
If necessary, this treatment can be repeated.

Method of Administration:
After addition of the solvent to the freeze-dried substance, the reconstituted solution should be slowly administered intramuscularly or subcutaneously.

Contraindications

- Hypersensitivity to human gonadotrophins or any of the excipients of Pubergen.
- Presence of uncontrolled non-gonadal endocrinopathies (e.g. thyroid, adrenal or pituitary disorders).
- Breast, uterine, ovarian tumours.
- Vaginal bleeding of unknown cause.
- Known or suspected androgen-dependent tumours such as testicular tumours, carcinoma of the prostate or mammary carcinoma in males.
- Malformations of the sexual organs incompatible with pregnancy.
- Fibroid tumours of the uterus incompatible with pregnancy.

Warnings and Precautions:
In males and females:
Hypersensitivity reactions: Hypersensitivity reactions, both generalized and local; anaphylaxis; and angioedema have been reported. If a hypersensitivity reaction is suspected, discontinue **PUBERGEN** and assess for other potential causes for the event. The ovulation cycle should be monitored with oestrial levels and ultrasonography.

General:
Patients should be evaluated for uncontrolled non-gonadal endocrinopathies (e.g. thyroid, adrenal or pituitary disorders) and appropriate specific treatment given. **PUBERGEN** should not be used for body weight reduction. HCG has no effect on fat metabolism, fat distribution or appetite.

In the female:
Multi-fetal gestation and birth: In the pregnancies occurring after induction of ovulation with gonadotrophic preparations, there is an increased risk of multiple pregnancies.

Ectopic pregnancy:
Infertile women undergoing Assisted Reproductive Technologies (ART) have an increased incidence of ectopic pregnancy. Early ultrasound confirmation that a pregnancy is intrauterine is therefore important.

Pregnancy Loss :
Rates of pregnancy loss in women undergoing ART are higher than in the normal population.

Congenital Malformations:
The incidence of congenital malformations after Assisted Reproductive Technologies(ART) may be higher than after spontaneous conceptions.This slightly higher incidence is thought to be related to differences in parental characteristics (e.g., maternal age, sperm characteristics) and to the higher incidence of multiple gestations after ART. There are no indications that the use of gonadotrophins during ART is associated with an increased risk of congenital malformations.

Ovarian Hyperstimulation Syndrome (OHSS):
OHSS is a medical event distinct from uncomplicated ovarian enlargement. Clinical signs and symptoms of mild and moderate OHSS are abdominal pain, nausea, diarrhea, mild to moderate enlargement of ovaries and ovarian cysts. Severe OHSS may be life-threatening. Clinical signs and symptoms of severe OHSS are large ovarian cysts, acute abdominal pain, ascites, pleural effusion, hydrothorax, dyspnea, oliguria, hematological abnormalities and weight gain. In rare instances, venous or arterial thromboembolism may occur in association with OHSS. Transient liver function test abnormalities suggestive of hepatic dysfunction with or without morphologic changes on liver biopsy have also been reported in association with OHSS.
OHSS may be caused by administration of human Chorionic Gonadotropin (hCG) and by pregnancy (endogenous hCG). Early OHSS usually occurs within 10 days after hCG administration and may be associated with an excessive ovarian response to gonadotropin stimulation. Late OHSS occurs more than 10 days after hCG administration, as a consequence of the hormonal changes with pregnancy. Because of the risk of developing OHSS, patients should be monitored for at least two weeks after hCG administration.

Women with known risk factors for a high ovarian response may be especially prone to the development of OHSS during or following treatment with **PUBERGEN**. For women having their first cycle of ovarian stimulation, for whom risk factors are only partially known, close observation for early signs and symptoms of OHSS is recommended.

To reduce the risk of OHSS, ultrasonography assessments of **follicular** development should be performed prior to treatment and at regular intervals during treatment. The concurrent determination of serum estradiol levels may also be useful. In ART, there is an increased risk of OHSS with 18 or more **follicles** of 11 mm or more in diameter. When there are 30 or more **follicles** in total, it is advised to withhold hCG administration.

Depending on the ovarian response, the following measures can be considered to reduce the risk of OHSS:

- withhold further stimulation with a gonadotropin for a maximum of 3 days (coasting);
- withhold hCG and cancel the treatment cycle;
- administer a dose lower than 10,000 IU of urinary hCG for triggering final oocyte maturation, e.g., 5,000 IU urinary hCG or 250 micrograms rec-hCG (which is equivalent to approximately 6,500 IU of urinary hCG);
- cancel the fresh embryo transfer and cryopreserve embryos;
- avoid administration of hCG for luteal phase support.

Adherence to the recommended **PUBERGEN** dose and treatment regimen and careful monitoring of ovarian response is important to reduce the risk of OHSS. If OHSS develops, standard and appropriate management of OHSS should be implemented and followed.

Ovarian torsion:
Ovarian torsion has been reported after treatment with gonadotrophins, including **PUBERGEN**. Ovarian torsion may be related to other conditions, such as OHSS, pregnancy, previous abdominal surgery, past history of ovarian torsion, and previous or current ovarian cysts. Damage to the ovary due to reduced blood supply can be limited by early diagnosis and immediate detorsion.

Vascular Complications:
Thromboembolic events, both in association with and separate from OHSS, have been reported following treatment with gonadotrophins, including **PUBERGEN**. Intravascular thrombosis, which may originate in venous or arterial vessels, can result in reduced blood flow to vital organs or the extremities. Women with generally recognised risk factors for thrombosis, such as a personal or family history, severe obesity or thrombophilia, may have an increased risk of venous or arterial thromboembolic events, during or following treatment with gonadotrophins. In these women the benefits of IVF treatment need to be weighed against the risks. It should be noted, however, that pregnancy itself also carries an increased risk of thrombosis.

In the male:
Antibody formation: Administration of hCG can provoke the formation of antibodies against hCG. In rare cases, this may result in an ineffective treatment. Treatment with hCG leads to increased androgen production. Therefore Patients with latent or overt cardiac failure, renal dysfunction, hypertension, epilepsy or migraine (or a history of these conditions) should be kept under close medical supervision, since aggravation or recurrence may occasionally be induced as a result of increased androgen production. HCG should be used cautiously in prepubertal boys to avoid premature epiphyseal closure or precocious sexual development. Skeletal maturation should be monitored regularly.

Interaction of other medicaments:
No interaction studies have been performed; interactions with commonly used medicinal products can therefore not be excluded. Following administration, human chorionic gonadotropin may interfere for upto ten days with the immunological determination of serum/urinary human chorionic gonadotropin, leading to a false positive pregnancy test.

Pregnancy and lactation:
PUBERGEN may be used for luteal phase support only, but must not be used afterwards during pregnancy. It must not be used during lactation.

Side Effects:
Frequency is unknown for all undesirable effects described below (cannot be determined with available data).

Immune system disorders
In rare cases generalized rash or fever may occur.

General disorders and administrative site conditions
Local site reactions such as bruising, pain, redness, swelling and itching, Oedema. Occasionally allergic reactions have been reported, mostly manifesting as pain and/or rash at the injection site. Tiredness.

Nervous system disorders
Headache.

Psychiatric disorders
Mood changes.

In the female
Reproductive system and breast disorders
Unwanted ovarian hyperstimulation, mild or severe ovarian hyperstimulation syndrome (OHSS, Warning and Precautions):
Mild OHSS: Painful breasts
Mild to moderate enlargement of ovaries

For the use of a Registered Medical Practitioner, Hospital or Laboratory

PUBERGEN 7500 IU
Chorionic Gonadotropin
For Injection USP 7500 IU/ Vial
For IM/SC use only

Composition:
Each vial of sterile freeze-dried product contains:
Human Chorionic Gonadotropin USP 7500 IU

Reconstitute with 1mL of 0.9% w/v sodium chloride Injection USP provided in this pack

Description:
Product: A White or almost white freeze-dried cake / powder sealed in a 2 mL USP Type -I clear glass vial.

Reconstitution solution: A clear, colourless solution is filled and duly sealed in a 2 mL USP Type-I clear glass ampoule.

After reconstitution: Clear colourless solution.

The product **PUBERGEN** is compatible with the reconstitution solution i.e. 0.9% W/V sodium chloride injection.

Pharmacological Information
Pharmacodynamic properties:
Pharmacotherapeutic group: Gonadotropins
ATC code : G03GA01
PUBERGEN is a preparation of human chorionic gonadotropin obtained from the urine of pregnant women. It stimulates the steroidogenesis in the gonads by virtue of a biological effect similar to that of LH (Luteinizing hormone, which is the same as interstitial cell stimulating hormone). In the male it promotes the production of testosterone and in the female the production of estrogens and particularly of progesterone after ovulation. In certain cases, this preparation is used in combination with human menopausal gonadotrophin (HMG).
Because Human Chorionic Gonadotropin is of human origin, no antibody formation is to be expected.

Pharmacokinetic properties :
Maximal hCG plasma levels will be reached in males approximately six and sixteen hours after a single IM or SC injection of hCG respectively, and in females after approximately 20 hours. Although high intersubject variability was observed, the difference related to gender after IM injection may be caused by gluteal fat thickness in women which exceeds that in men. HCG is for approximately 80 per cent metabolized, predominantly in the kidneys. IM and SC administration of hCG were found to be bioequivalent regarding the extent of absorption and the apparent elimination half-lives of approximately 33 hours. On basis of the recommended dose regimens and elimination half-life, cumulation is not expected to occur.

Indication:
In female:

- Ovulation induction in sub fertility due to anovulation or impaired follicle-ripening.
- Preparation of follicles for puncture in controlled ovarian hyperstimulation programs (for medically assisted reproductive techniques).
- Luteal phase support.

In male:

- Hypogonadotropic hypogonadism (also cases of idiopathic dysspermias have shown a positive response to gonadotropins).
- Delayed puberty associated with insufficient gonadotropic pituitary function.
- Cryptorchidism not due to anatomical obstruction.

Dosage and administration:
Dosage in the female:

- Ovulation induction in subfertility due to anovulation or impaired follicle-ripening**
Usually, one injection of 5000-10000 IU to complete treatment with an FSH-containing preparation
- Preparation of follicles for puncture in controlled ovarian hyperstimulation programs**
Usually, one injection of 5000–10000 IU to complete treatment with an FSH-containing preparation.
- Luteal phase support**
Two to three repeat injections of 1000 to 3000 IU may be given within 9 days following ovulation or embryo transfer (for example on day 3, 6 and 9 after ovulation induction) .

Dosage in the male:

- Hypogonadotropic hypogonadism**
1000 - 2000 IU two to three times per week. If the main complaint is subfertility, may be given with an additional follitropin (FSH)-containing preparation two to three times a week. This treatment should be continued for at least three months before any improvement in spermatogenesis can be expected. During this treatment testosterone replacement therapy should be suspended. Once achieved, the improvement may

Ovarian cysts
Abdominal pain
Abdominal discomfort
Gastrointestinal symptoms such as nausea, diarrhoea and bloating
Severe OHSS: Large ovarian cysts (prone to rupture),
Acute abdominal pain
Ascites
Weight gain
Hydrothorax In rare instances, thromboembolism has been associated with FSH/human chorionic gonadotropin therapy

Not all symptoms described are always associated to OHSS.

In the male
Metabolism and nutrition disorders
Water and sodium retention is occasionally seen after administration of high dosages; this is regarded as a result of excessive androgen production.

Reproduction system and breast disorders
Human chorionic gonadotropin treatment may sporadically cause gynaeomastia.

Skin and subcutaneous tissue disorders
Acne may occur occasionally during human chorionic gonadotropin therapy.

Overdose and treatment:
The toxicity of human chorionic gonadotrophic hormone is very low. However, too high a dose may lead to hyperstimulation of the ovaries

Instructions for Usage:

- The powder for injection is reconstituted by adding the solvent.
- Do not use if the solution contains particles or if the solution is not clear. The solution should be used immediately after reconstitution.
- Any unused product or waste material should be disposed of in accordance with local requirements

Effects on ability to drive and use machines:
As far as Known **PUBERGEN** has no influence on the ability to drive and use machines.

Pre Clinical Safety Data: There are no preclinical data of relevance to the prescriber which are additional to that already included in other sections of the SPC.

Storage conditions:
Storage condition of the product PUBERGEN 750 0 IU:Store in a refrigerator (2°C to 8°C). Store in the original carton to protect from light. Do not freeze.

Storage condition of the diluent: Store in a refrigerator (2°C to 8°C). Store in the original carton to protect from light. Do not freeze.

Storage condition of the reconstitution solution: After reconstitution, solution should be used immediately and discard the unused solution.

Shelf life:
Shelf life of the product PUBERGEN 7500 IU: 3 years when stored at a temperatures between 2°C – 8°C.

Shelf life of the diluent: 5 years when stored at a temperatures between 2°C – 8°C.

Shelf life of reconstitution solution: After reconstitution, solution should be used immediately and discard the unused solution.

Incompatibilities :
In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

Dosage forms and packing available:
Box containing 1 Vial of sterile freeze dried Chorionic Gonadotropin for Injection USP + 1 Ampoule of Sodium Chloride Injection USP (0.9% w/v).

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Product Registration Holder:
Healol Pharmaceuticals Sdn. Bhd.,
74-3, Jalan Wangsa Delima 6, KLSC,
Wangsa Maju, 53300, Kuala Lumpur, Malaysia.

Manufactured by: Sanzyme(P) Ltd.
Plot No. 8, Sy. No. 542, Phase-II, Koltur (V),
Shamirpet (M), Medchal-Malkajgiri (D),
500 101, Telangana State, INDIA.

210 mm

140 mm